

DARABANI, Romania

WMO: 150000

Lat: 48.1950N

Lon: 26.5736E

Elev: 854

StdP: 14.25

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.6	8.4	0.8	5.9	5.6	5.0	7.3	9.3	31.0	25.5	27.1	24.5	7.3	320	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.8	88.4	67.5	85.3	67.1	82.6	66.1	71.2	81.9	69.5	81.0	68.1	78.8	6.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.0	106.0	76.3	66.2	99.5	72.5	64.6	94.2	71.5	35.6	80.8	34.0	81.8	32.8	78.8	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.1	19.0	17.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	25.1	32.3	39.0	49.9	60.8	67.6	69.4	70.5	61.3	49.3	39.0	30.0
	DBStd	17.58	9.09	9.53	9.07	8.94	6.86	5.88	5.38	6.32	8.24	7.89	8.57	9.53
	HDD50	2827	773	497	350	116	9	0	0	0	7	110	344	621
	HDD65	6178	1238	916	806	455	160	41	20	21	167	488	780	1086
	CDD50	2678	0	0	9	112	346	527	600	635	346	89	14	0
	CDD65	552	0	0	0	1	31	117	155	191	56	1	0	0
	CDH74	4350	0	0	0	17	240	815	1132	1673	467	6	0	0
	CDH80	1253	0	0	0	0	32	173	318	613	117	0	0	0
Wind	WSAvg	7.6	8.0	8.6	8.4	9.2	7.8	6.9	6.2	6.6	7.1	7.2	8.1	7.7
Precipitation	PrecAvg	29.3	1.8	1.9	1.9	2.0	3.0	3.8	3.9	2.8	2.4	1.9	1.8	2.0
	PrecMax	39.8	3.0	3.7	4.3	5.5	6.2	8.0	8.2	7.6	5.5	5.9	3.3	4.6
	PrecMin	20.6	0.6	0.9	0.6	0.5	1.3	1.6	1.6	0.9	0.5	0.1	0.0	0.5
	PrecStd	4.3	0.7	0.7	0.9	0.9	1.3	1.5	1.9	1.5	1.4	1.1	0.7	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.2	56.6	66.6	76.4	83.7	86.9	90.6	93.6	89.1	74.2	62.8	53.4
		MCWB	41.6	48.4	52.9	58.2	63.4	70.3	68.9	68.6	68.3	60.7	52.8	46.8
	2%	DB	42.0	51.2	59.6	72.4	79.8	84.2	86.9	89.6	82.8	69.3	57.8	49.0
		MCWB	39.2	45.2	48.6	56.8	62.7	69.4	68.2	66.8	64.4	58.6	50.7	44.9
	5%	DB	39.1	46.7	55.1	68.3	76.5	81.8	83.5	86.3	79.0	65.3	53.9	45.5
		MCWB	36.9	42.0	46.3	55.0	61.6	67.1	67.9	66.3	63.0	56.4	48.4	42.5
	10%	DB	36.3	43.8	50.8	64.6	73.1	79.0	80.1	83.1	75.3	61.0	50.5	41.4
		MCWB	34.3	40.5	43.9	53.5	59.7	65.8	66.3	65.4	62.1	54.5	46.5	39.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.8	49.9	53.5	60.5	67.5	75.7	72.9	71.4	69.1	62.8	54.0	48.0
		MCDB	44.5	56.2	64.2	71.1	77.8	82.6	84.2	82.2	86.7	68.5	59.9	51.2
	2%	WB	39.7	45.6	50.2	58.3	65.3	71.7	71.3	69.6	66.6	59.9	51.3	44.8
		MCDB	41.3	50.3	57.8	68.8	74.5	79.0	83.1	83.1	78.6	67.0	55.7	48.1
	5%	WB	36.9	43.1	47.5	55.7	63.3	69.1	69.6	68.3	64.6	57.8	48.3	42.2
		MCDB	38.5	46.4	54.3	64.6	72.3	78.2	80.8	80.4	74.7	63.6	51.2	44.9
	10%	WB	34.5	40.7	45.0	53.6	61.2	67.1	68.0	67.0	63.0	55.4	45.7	39.3
		MCDB	35.7	43.5	49.9	62.1	70.2	75.6	77.9	77.6	73.3	59.9	48.3	41.0
Mean Daily Temperature Range	5% DB	MDBR	8.8	10.9	13.7	17.1	17.6	17.1	17.5	18.8	16.2	13.4	9.2	8.4
		MCDBR	11.7	16.9	21.2	23.7	22.1	20.6	22.0	22.8	22.6	20.4	14.5	13.8
		MCWBR	9.2	11.6	12.6	11.3	9.3	9.1	8.1	7.8	8.7	10.1	9.6	10.3
	5% WB	MCDBR	11.3	16.1	19.4	22.0	20.1	18.9	20.9	20.0	20.7	17.2	13.6	12.5
		MCWBR	9.2	11.5	11.7	11.2	9.5	9.4	8.7	7.6	8.4	9.3	9.5	10.0
Clear-Sky Solar Irradiance	taub	0.313	0.343	0.380	0.436	0.409	0.408	0.437	0.442	0.400	0.366	0.340	0.311	
	taud	2.277	2.232	2.228	2.113	2.264	2.294	2.220	2.200	2.275	2.349	2.358	2.338	
	Ebn at Noon	234	253	262	258	271	272	261	253	253	242	223	221	
	Edh at Noon	24	31	37	46	41	40	43	42	35	28	23	20	
All-Sky Solar Radiation	RadAvg	338	585	974	1407	1789	1892	1856	1659	1157	727	336	252	
	RadStd	52	91	104	134	147	152	126	118	154	96	32	36	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	+1.26	N/A	N/A	+1.55	N/A	N/A	-265	-380	+189	+68

Nomenclature: See separate page